



# **Team 15 Sprint 2 Retrospective**

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## What Went Well?

We were able to improve from the execution of the last sprint, and partially implement every user story that was listed. Along with that, we had more meetings as a group, and were able to communicate progress and issues that were occurring as we continued development. It was not perfect, but there were a lot of strides we made that made us feel like we accomplished much more than we did in the first sprint.

## User Story 11

| User Story # | Description                                                                                                                                       |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 11           | As a dieter and athlete, I would be able to switch between imperial and metric systems to measure weights in the gym and portions in the kitchen. |

**Completed:** Imperial and metric conversions, found in the settings tab of the app, have been fully implemented. Users are able to select their preference, which is stored in the app, so users do not have to reselect it. Stored workouts and diets are converted based on the selected measuring system, and suggested workouts are changed based on what the user prefers for measurements.

## User Story 14

| User Story # | Description                                                                                    |
|--------------|------------------------------------------------------------------------------------------------|
| 14           | As a user, I would like to be able to select a location, so I can see gyms and people near me. |

**Completed:** This was an unfinished user story from Sprint 1. While in theory it was functional, it wasn't testable, as our mapping library only worked on mobile, and the emulator being an emulator didn't have a physical location. This sprint was successfully implemented on the web. However, the CSS was modified before the presentation, causing it to break. Overall, mostly successful, but required a little bit more extensive testing.

## User Story 15

| User Story # | Description |
|--------------|-------------|
|--------------|-------------|

|    |                                                                                                      |
|----|------------------------------------------------------------------------------------------------------|
| 15 | As a user, I would like to be able to enable certain accessibility features (dark mode, large text). |
|----|------------------------------------------------------------------------------------------------------|

**Completed:** This user story was left over from Sprint 1. The actual features were fully implemented, but the settings selected by a user were not saved between app restarts. This was finished in this Sprint, where when a user chooses their preferred accessibility settings, they are now saved in the database, and automatically loaded and reapplied when the app is restarted.

## User Story 18

| User Story # | Description                                                                                              |
|--------------|----------------------------------------------------------------------------------------------------------|
| 18           | As a dieter, I would be able to view the nutritional facts of different meals from national restaurants. |

**Completed:** This was leftover from Sprint 1, picking up after I found and cleaned restaurant menu data online, and added a pipeline to derive features. This time around, I finalized the UI functionalities and properly routed it to the backend. Now, users are fully able to filter-search meals from national chain restaurants and add them to their log.

## User Story 19

| User Story # | Description                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| 19           | As a user, I will be able to bypass manually inputting username and password during login with a Remember Me feature |

**Completed:** This was leftover from Sprint 1, picking up after all the backend functions and server logic were implemented but no frontend. Improving on this, I properly debugged and connected the client and server to ensure smooth functionality. Now, users are fully able to bypass the login screen if they toggle the Remember Me login feature.\

## User Story 20

| User Story # | Description |
|--------------|-------------|
|--------------|-------------|

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|----|--------------------------------------------------------------------------------------------------|
| 20 | As a user, I would like to be able to read and agree to the Terms and Conditions of the service. |
|----|--------------------------------------------------------------------------------------------------|

**Completed:** This was leftover from Sprint 1, but was relatively simple to implement. Using a PDF library for mobile and a simple iframe for web, I was able to embed the terms and conditions PDF file into a screen before onboarding. The server was slightly tweaked to take in a “tocAccepted” parameter on account creation, and without this flag, account creation was denied. Users are prompted to agree to terms and conditions with a checkbox, and failing to comply will result in them not moving forward in onboarding with an error.

## User Story 21

| User Story # | Description                                                                              |
|--------------|------------------------------------------------------------------------------------------|
| 21           | As a user, I would like to be able to set up app notifications for workouts, meals, etc. |

**Completed:** Using the expo-notifications library, I was able to implement the feature for app notifications. In the settings, users can enable/disable notifications for workouts and meals. For meals, the user selects their desired time for breakfast/lunch/dinner and will receive a reminder notification at the specified time. For workouts, a user will add a workout on the calendar, and then receive a notification at that day/time.

## User Story 24

| User Story # | Description                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 24           | As a user, I want to generate a quick meal recommendation so that I can easily decide what to eat based on my preferences and goals. |

**Completed:** When the user requests a quick meal, the system retrieves relevant meal options from the database using the user’s profile data (such as preferences, dietary restrictions, and goals). These results are then formatted into a structured prompt along with the user’s profile information and passed to the LLM. The LLM generates a personalized quick meal recommendation based on this context. The response is then displayed to the user in the interface. This allows users to quickly get tailored meal suggestions without manually searching or filtering through options.

## User Story 25

| User Story # | Description                                                                    |
|--------------|--------------------------------------------------------------------------------|
| 25           | As a dieter, I will be able to chart intake and set goals for macro and water. |

**Completed:** When a user navigates to the diet tab, they are able to view multiple bars, which represent different macros that the user might want to have tracked. Users are able to input their goals, and then they can click on the bar to log or double-click to remove a certain amount from the tracking. In addition to this, users can also have their trackers populated automatically if they choose to assign macro values to the meals that they add, which will be expanded on in user story 28.

## User Story 26

| User Story # | Description                                                                                  |
|--------------|----------------------------------------------------------------------------------------------|
| 26           | As a new user, I will have my day-one caloric goals displayed based on onboarding responses. |

**Completed:** During the onboarding questionnaire, users input personal information including their age, gender, weight, and height. Leveraging these, we calculate the Basal Metabolic Rate, which is the profile's target needed calories, from the Mifflin-St Jeor formula and write it to the database. This attribute is then read and displayed in the Calorie Goals widget on the diet tab.

## User Story 27

| User Story # | Description                                                                                    |
|--------------|------------------------------------------------------------------------------------------------|
| 27           | As a dieter, I will press a button to recalibrate my caloric goals based on recorded behavior. |

**Completed:** As users log a bunch of activity, their originally calculated day-1 calorie goals become outdated with respect to their health goals. Pressing the Recalibrate button pulls the profile and their workout & diet history from the database, formats into a string prompt, and calls the LLM to ask if the current goal should change. If changed, we write over the current in the

database, which updates the display on the diet tab. Ideally, this would be pressed only once every morning by users so they can plan their day around it.

## User Story 28

| User Story # | Description                                                                              |
|--------------|------------------------------------------------------------------------------------------|
| 28           | As a user, I would be able to assign macros and ingredients to my meals when I add them. |

**Completed:** Building on the tagging system made in Sprint 1, the user is now able to assign macros and ingredients to meals that they make at home. In order to not make the UI too long, the tagging and macro sections are nestled in dropdown menus, and the ingredients section is a different tab in the add tag meal section. In the browsing section, users are once again greeted with similar dropdown menus to filter either by tags, macros, or ingredients. The cards themselves display all of the relevant information, with the option to expand and show more data in the form of the actual ingredients. Along with this, any meals with macros assigned to it will automatically populate the tracker developed in user story 25.

## User Story 30

| User Story # | Description                                                                                                                                |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 30           | As a gym user, I can have my weight, sets, and reps tailored for a specific ongoing exercise, accounting for all necessary logged progress |

**Completed:** While a user is logging a gym session, they are able to press the Tailor button to calibrate the weight, sets, and reps for the current exercise in their split routine. We pull the profile and workout & diet history from the database, format it into a string prompt, and call the LLM to optimize their weight/sets/reps, then display it in the logging text fields. Ideally, this would be used by experienced athletes that know which exercises they like but don't know the optimal weight/sets/reps for their routine and physique.

## User Story 31

| User Story # | Description |
|--------------|-------------|
|--------------|-------------|

|    |                                                                                                                                                   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 31 | As a gym user, I will have access to a help feature that suggests alternative options if a machine or exercise isn't available or is undesirable. |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------|

**Completed:** When a user uses our LLM generated workout feature, each exercise the LLM generates has an “alternatives” button attached. Pressing this button prompts the LLM once more with information about the user alongside the exercise they are requesting to modify, and asks the LLM for a JSON object containing alternatives we can format. We then format the alternatives in a text list display for the user. This is helpful if the user has an injury, or doesn't have access to the exercise/machine at their local gym.

## User Story 32

| User Story # | Description                                                                                                      |
|--------------|------------------------------------------------------------------------------------------------------------------|
| 32           | As a beginner in my fitness journey, I will have access to a help feature that provides assistance on exercises. |

**Completed:**

If a user is new to the gym and does not know how to do a certain exercise they can use this assistance feature to receive guidance. To do so, they will navigate to the last tab and click the exercises button, where they will see a list of all the available exercises. Upon clicking the help button for a specific exercise, the server will retrieve detailed step by instructions from the database and display it to the user.

## User Story 33

| User Story # | Description                                                                               |
|--------------|-------------------------------------------------------------------------------------------|
| 33           | As a gym user, I would like to view a graph of weight progression for a specific exercise |

**Completed:** Whenever a user looks back at a workout they have logged, each exercise has a “progression” button next to it. When the user clicks this button, we pull their previously logged data from their profile, and feed it into an LLM. We ask the LLM given the user's profile and previous data, what would be reasonable data to expect in the future? Then we plot the users previous data, and future data, against each other on a graph. This is extremely helpful for users to make sure they're making progress, look back at their old progress, and also know what to reasonably expect in terms of future progress so they have reasonable goals to hit and attain.

## What Did Not Go Well?

We still struggled to completely finish a few user stories, and there was a lot of time spent on debugging issues at the end of the sprint, which we could have used to finish these stories. Also, communication of implementations between teammates was not perfect, as we had a couple of issues that would have been much simpler if communication had been clearer. Time allocation was still not perfect and could have been improved.

### User Story 15

| User Story # | Description                                                                                       |
|--------------|---------------------------------------------------------------------------------------------------|
| 29           | As a user, I would like to be able to have weekly and monthly diet and workout reports generated. |

**Not Completed:** The temporary weekly and monthly reports that were generated were what was expected, generating a value for the given specifics. The only unimplemented feature was connecting it to each user. Placeholder data was used, assuming the user had never opened the app before, so connecting it to generate accurate data for a specific user is what was not completed accurately in this sprint.

### User Story 22

| User Story # | Description                                                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 22           | As a gym user, I will be able to generate an all-inclusive, tailored, generic Quick Workout, accounting for every possible profile characteristic. |

**Not Completed:** Workouts are generated based on users' profiles, meaning the general user story was implemented. However, specific acceptance criteria were not met, such as being able to edit the generated workout and having it logged for users if they select to. Although the general idea of the user story is completed, the details and specifications we wanted are not fully finished.

### User Story 23

| User Story # | Description                                                                                           |
|--------------|-------------------------------------------------------------------------------------------------------|
| 23           | As a gym user, I will be able to generate a tailored workout purely based on a specific muscle group. |



**Not Completed:** This user story was almost completely implemented; there was just one acceptance criterion that was not met. If a muscle group was not selected, it still generated a workout, so fixing that bug is the step to make this user story fully implemented.

## **How Should We Improve?**

### **Even more frequent communication and check-ins**

No more pushing entire user stories the day / minutes before we present.

### **Before pushing, ensure functionality and make tests**

Make sure that whatever we push is 100% functionable to the best of our ability. This doesn't mean we have to throw entire user stories into one push, but when we push something partially complete, make sure that the partially complete is actually correct.

Even better, if we push developer or manual tests with our features, it is way easier to debug and pinpoint flaws in the application when we add onto the features or change them with new functionalities.